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AT-1

M. Nandhini
192472318
CSE-AI

Short - Answer - Test

1. **Illumination and object properties:**
A bright light behind the object makes the background very bright and the object appear dark. Object reflectance absorption, and surface properties affect the reflected light

2. **Different brightness and contrast:**
The image formation model is: $f(x,y) = i(x,y) \times r(x,y)$
where i is illumination and r is reflectance.
Different lighting, exposure, aperture, and sensor response can produce different brightness and contrast.

3 **Computer vision in security:**
Computer vision analyses video using: Motion detection, Object detection, Object tracking, Activity recognition.
Video → Processing → Detection → Tracking → Suspicious activity → Alert.

4 **OCR accuracy:**
Good image acquisition requires proper resolution, focus, lighting and exposure. Preprocessing such as grayscale conversion, noise removal, thresholding, and skew correction improves character recognition.
Good image → Preprocessing → clear characters → Better OCR.

5 **Intensity resolution:**
Intensity resolution is the ability to distinguish small differences in brightness. In medical images, tissues may have similar intensities.

6 Sampling and aliasing:

Sampling convets a continuous image into digital pixels. If the sampling rate is too low for fine repetitive Patterns, aliasing occurs.

Low Sampling → under Sampling → Aliasing

7 Satellite image improvement.

Small buildings require high spatial resolution. Improvement can be achieved by: increasing sensor resolution. Reducing GSD, improving optics, increasing sampling rate, Higher spatial resolution → more details → Better building identification

8 Robot in warehouse:

The camera first senses and captures the image. Image processing improves it, and computer vision detects and locates objects.

Sensing → Processing → Object detection → location → Robot action.

9 Industrial inception under different lighting.

Different illumination changes the intensity and apperance of same object. Camera expose, shadow, reflections during image acquisition can therfore cause inconsistence inspection.

10 wild life monitoring:

Cameras acquire image / videos of animals. Digital representation stores these images as pixels & intensity / color values, while computer vision detects, identifies & tracks animals automatically.